Sequence 14

# PROBABILITY: CONDITIONING AND WEIGHTED TREES

26 April–24 May 2027

## I Sample space, outcomes and events

### Vocabulary

A random experiment has several possible results, without it being known in advance which will occur. An outcome is a possible result; the sample space $\Omega$ is the set of them. An event is a subset of $\Omega$; an elementary event contains a single outcome. The impossible event is $\varnothing$; the certain event is $\Omega$.

### Operations on events

$\overline{A}$ is the complementary event of $A$. $A \cap B$ means "$A$ and $B$"; $A \cup B$, "$A$ or $B$", possibly both. Two events are mutually exclusive if $A \cap B = \varnothing$; otherwise, they are compatible.

**Example 14.1**

A six-sided die is rolled. Write $\Omega$, $A$: "get an even number", $B$: "get 4", then $\overline{A}$, $A \cap B$ and $A \cup B$. Are $A, B$ compatible?

**Solution:**

$\Omega = \{1;2;3;4;5;6\}$, $A = \{2;4;6\}$, $B = \{4\}$, $\overline{A} = \{1;3;5\}$, $A \cap B = \{4\}$, $A \cup B = A$. The events are compatible because they share the outcome 4.

**Example 14.2**

With $A = \{a;b;c;d;e\}$ and $B = \{b;e;f;g;h\}$, give the intersection and the union. Can $\overline{A}$ be determined without specifying the sample space?

**Solution:**

$A \cap B = \{b;e\}$ and $A \cup B = \{a;b;c;d;e;f;g;h\}$. The complement depends on the sample space; it cannot be determined without this information.

## II Probability model and equally likely outcomes

**Probability on a finite sample space**

To each outcome, a model assigns a probability that is positive or zero; the sum is 1. The probability of an event is the sum of the probabilities of its outcomes. We have

$$P(\Omega) = 1, \quad P(\varnothing) = 0, \quad P(\overline{A}) = 1 - P(A).$$

With equally likely outcomes, all outcomes have the same probability and

$$P(A) = \frac{\text{Card}(A)}{\text{Card}(\Omega)}.$$

Equal likelihood is an assumption of the model, not a consequence of the list of outcomes alone.

**Example 14.3**

Assume the die is fair. For $A =$ "even" and $C =$ "greater than or equal to 5", calculate $P(A)$, $P(C)$, $P(A \cap C)$ and $P(A \cup C)$.

**Solution:**

$P(A) = 3/6 = 1/2$, $P(C) = 2/6 = 1/3$, $A \cap C = \{6\}$ so $P(A \cap C) = 1/6$. $A \cup C = \{2; 4; 5; 6\}$, so $P(A \cup C) = 4/6 = 2/3$. The outcome 6 must not be counted twice.

**Example 14.4**

A drawing pin can land point up or point down. Can we deduce that each outcome has probability 1/2? How can a model be proposed?

**Solution:**

No: two outcomes are not enough to impose equal likelihood. One can repeat the throws under comparable conditions and use the observed frequencies to propose a model, bearing in mind fluctuations and the characteristics of the object.

**Teacher notes**

Reactivate the link between observed frequency and postulated probability. An experimental result does not prove exact equal likelihood.

## III Conditional probability from a table

**Definition**

For $P(A) > 0$, the probability of $B$ given $A$ is

$$P_A(B) = \frac{P(A \cap B)}{P(A)}, \qquad P(A \cap B) = P(A)P_A(B).$$

For an equally likely draw within a finite population:

$$P_A(B) = \frac{\text{Card}(A \cap B)}{\text{Card}(A)}.$$

We also write $P(B \mid A)$. $P_A(B)$ and $P_B(A)$ are generally not equal.

**Example 14.5**

A student is drawn uniformly from the class of 32 students studied earlier. $G$: "boy", $N$: "takes CS".

| | $G$ | $\overline{G}$ | Total |
|---|---|---|---|
| $N$ | 12 | 9 | 21 |
| $\overline{N}$ | 6 | 5 | 11 |
| Total | 18 | 14 | 32 |

Calculate $P(G)$, $P(N)$, $P(G \cap N)$, $P_N(G)$ and $P_G(N)$.

**Solution:**

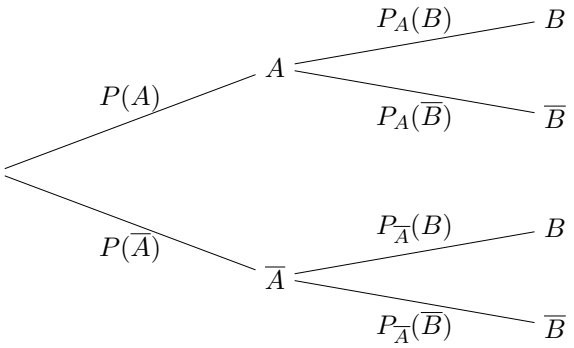
$P(G) = 18/32 = 9/16$, $P(N) = 21/32$, $P(G \cap N) = 12/32 = 3/8$. $P_N(G) = 12/21 = 4/7$, whereas $P_G(N) = 12/18 = 2/3$. By the formula: $P_N(G) = (12/32)/(21/32) = 12/21$. Given that the student takes CS, the reference population is reduced to the 21 students concerned.

**Watch out.** Conditioning requires the known event to have nonzero probability; we do not divide by zero.

## IV Weighted trees and the probability of a path

**Rules**

At each node, the sum of the probabilities of the outgoing branches is 1. A second-level branch carries a conditional probability. A path represents an intersection of events; its probability is the product of the probabilities written along that path.




**Example 14.6**

We know $P(A) = 0.6$, $P_A(B) = 0.7$ and $P_{\overline{A}}(\overline{B}) = 0.9$. Complete the other branches and calculate the probabilities of the four paths.

**Solution:**

$P(\overline{A}) = 0.4$, $P_A(\overline{B}) = 0.3$, $P_{\overline{A}}(B) = 0.1$.

$$P(A \cap B) = 0.6 \times 0.7 = 0.42, \quad P(A \cap \overline{B}) = 0.18,$$

$$P(\overline{A} \cap B) = 0.04, \qquad P(\overline{A} \cap \overline{B}) = 0.36.$$

The sum of the four probabilities is 1, which checks the consistency of the model.

**Example 14.7**

Justify the multiplication along the path $A$ then $B$ from the definition of $P_A(B)$.

**Solution:**

Since $P_A(B) = P(A \cap B)/P(A)$ and $P(A) > 0$, multiplying by $P(A)$ gives $P(A)P_A(B) = P(A \cap B)$.

## V From a table to a tree

**Example 14.8**

For the class of 32 students, construct a tree starting with CS / no CS, then boy / girl. Then construct the tree in the reverse order. Check, in both trees, the probability of choosing a girl who takes CS.

**Solution:**

> First tree: initial branches $P(N) = 21/32$, $P(\overline{N}) = 11/32$. After $N$: $P_N(G) = 12/21 = 4/7$, $P_N(\overline{G}) = 9/21 = 3/7$. After $\overline{N}$: 6/11 and 5/11. Second tree: initial branches $P(G) = 18/32 = 9/16$, $P(\overline{G}) = 14/32 = 7/16$. After $G$: $P_G(N) = 12/18 = 2/3$, then 1/3. After $\overline{G}$: $P_{\overline{G}}(N) = 9/14$, then 5/14. The required path is worth $(21/32)(9/21) = 9/32$ in the first tree and $(14/32)(9/14) = 9/32$ in the second. Changing the order preserves the probabilities of the intersections but changes the probabilities of the branches.

**Example 14.9**

An urn contains 3 red balls and 2 blue balls. Two balls are drawn in succession **without replacement**. Construct the tree of colours and calculate the probability of drawing two red balls.

**Solution:**

> First level: $R$ with 3/5, $B$ with 2/5. After $R$, 2 red and 2 blue remain: branches $1/2, 1/2$. After $B$, 3 red and 1 blue remain: branches $3/4, 1/4$. So $P(R_1 \cap R_2) = (3/5)(1/2) = 3/10$. Using 3/5 on the second red branch would ignore the absence of replacement.

## VI Inverting conditioning and false positives

**Example 14.10: Fictional component check**

A batch contains 10000 components, of which 100 are defective. A test is positive for 99 of the defective components and for 495 of the non-defective components. A component is chosen uniformly. $D$: "defective"; $T$: "test positive".

1. Complete the two-way table of test / true state.
2. Calculate $P_D(T)$ and $P_T(D)$.
3. Give the number of false positives and false negatives.
4. Calculate the sensitivity $P_D(T)$ and the specificity $P_{\overline{D}}(\overline{T})$.

**Solution:**

| | $D$ | $\overline{D}$ | Total |
|---|---|---|---|
| $T$ | 99 | 495 | 594 |
| $\overline{T}$ | 1 | 9405 | 9406 |
| Total | 100 | 9900 | 10000 |

$P_D(T) = 99/100 = 99\,\%$. On the other hand $P_T(D) = 99/594 = 1/6 \approx 16.67\,\%$. A positive test therefore does not imply here a 99 % probability of defect. The 495 false positives are the sound components that tested positive; the false negative is the defective component that tested negative. The sensitivity is 99 %; the specificity $9405/9900 = 95\,\%$. The low initial proportion of defective components explains why false positives outnumber true positives despite the good sensitivity.

**Reading a test result**

"Positive among the defective" and "defective among the positives" correspond to two different conditionings. Always name the reference group before the calculation.

**Teacher notes**

The numbers are fictional. The table allows studying the inversion without requiring a general total-probability formula: this is not an expectation of the 2026 Seconde programme.

## VII Simulation and the law of large numbers

**Popularised version**

When an experiment is repeated a large number of times in the same model with independent repetitions, the observed frequency of an event is, barring exceptions, close to its probability. It fluctuates: it does not necessarily get closer with each new trial.

```
from random import Random

def frequences_six(n, graine=1):
    # n entier >= 1 ; modele de lancers independants
    rng = Random(graine)
    succes = 0
    frequences = []
    for k in range(1, n+1):
        if rng.randint(1, 6) == 6:
            succes = succes+1
        frequences.append(succes/k)
    return frequences
```

**Example 14.11**

Plot the frequencies for $n = 1000$, with several seeds. Compare with $1/6$. Why fix the seed? Can we require the final frequency to be exactly $1/6$?

**Solution:**

The plots depend on the seed and fluctuate around $1/6$. Fixing the seed makes the simulation reproducible. With 1000 throws, a final frequency of exactly $1/6$ is even impossible, because $1000/6$ is not an integer. A nearby frequency is consistent with the model, but is not a proof of it.

**Example 14.12**

Modify the program to track the frequency of the event "even result". Which reference probability should be used?

**Solution:**

Replace the test with `rng.randint(1, 6) % 2 == 0`. The reference probability is $3/6 = 1/2$. Each trial must make a single draw; do not call the generator several times to test the same result.